1 - 31

2 - 31

3 - 34

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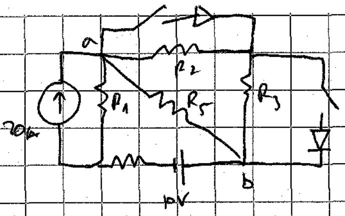
(96)

הערות הבודק

10 נק'

31/33

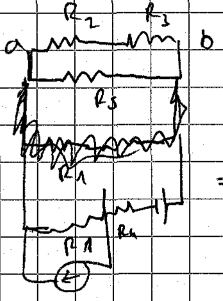
①

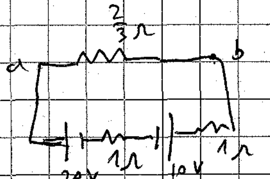

a
R_2
R_1
R_5
R_3
20A
10V
b

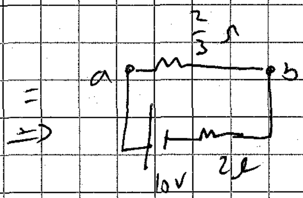
$R_1 = R_2 = R_3 = R_4 = R_5 = 1\Omega$

$V = 10V$

$$\frac{1}{\frac{1}{2}+\frac{1}{1}} = \frac{1}{\frac{1+2}{2}} = \frac{2}{3}$$


ב
a
b
R_2
R_3
R_5
R_1
R_4

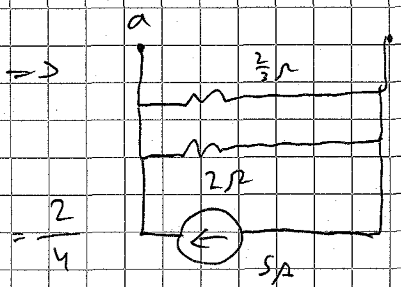

⇒
a
b
$\frac{2}{3}\Omega$
20V
1Ω
10V
1Ω

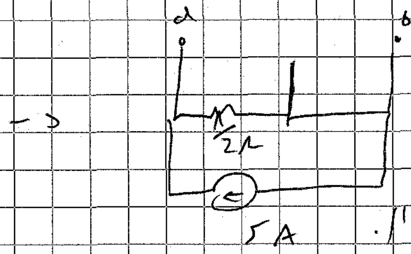

=
⇒
a
b
$\frac{2}{3}\Omega$
10V
2Ω

$10 = 2 \cdot I$

$I = 5$

$V = IR \quad 20 \cdot 1 = 20$

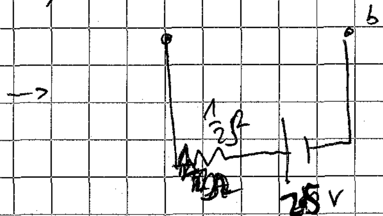

⇒
a
b
$\frac{2}{3}\Omega$
2Ω
5A


→
a
b
$\frac{1}{2}\Omega$
5A

שקול נורטון.

$$\frac{1}{\frac{3}{2}+\frac{1}{2}} = \frac{2}{4}$$

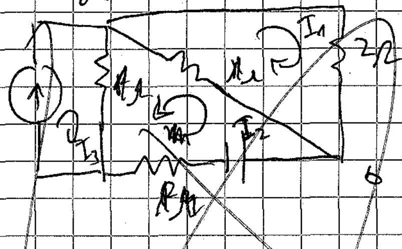
$5 \cdot 2 = 10$


→
b
$\frac{1}{2}\Omega$
2.5V

שקול תבנין

✓

②


a
I_1
2Ω
1Ω
1Ω
I_2
I_3
1Ω
b

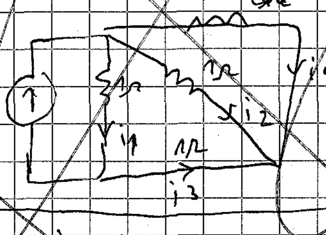
$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 10-20\cdot 1 \end{pmatrix}$$

$3I_1 + I_2 = 0 \qquad I_2 = -3I_1$

$I_1 + 3I_2 = -10 \qquad I_1 - 6I_1 = -10 \qquad -5I_1 = -10$

$I_1 = 2A$

$I_2 = -6A$


2Ω
1Ω
1Ω
i_4
i_2
i_1
1Ω
i_3

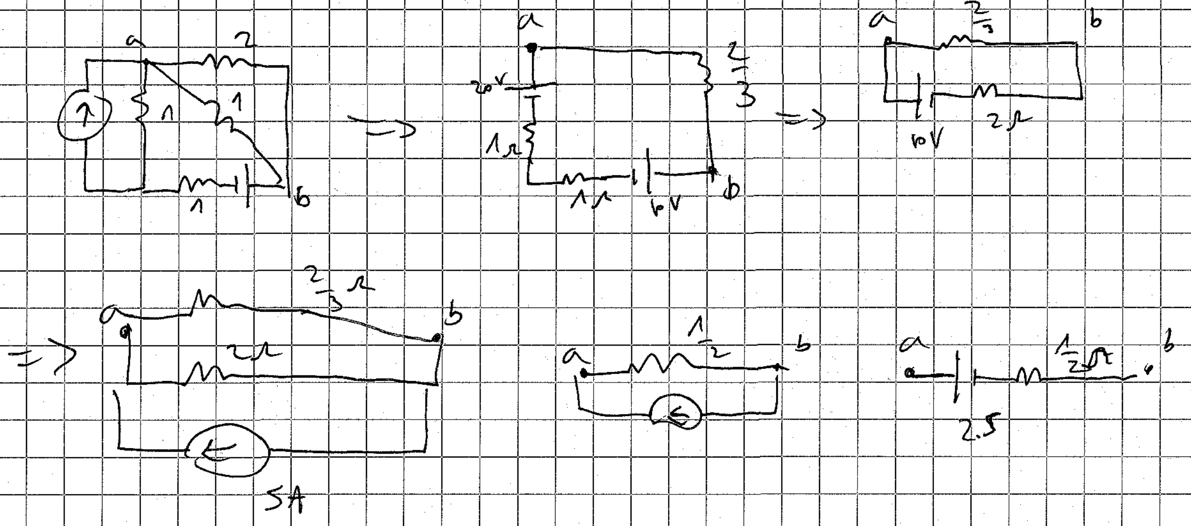
הסימון (לכיוון בציור הזה) ←

$$\begin{cases} i_1 = I_2 + 20 = 14A \\ i_2 = -I_1 + I_2 = +4A \end{cases}$$

$i_3 = I_2 = -6A \qquad i_4 = I_1 = 2A$

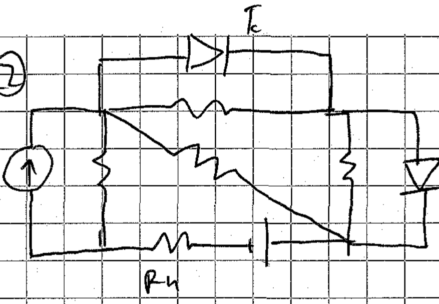
טיוטה למבחן

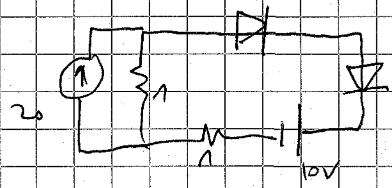
סע' 1


a
2
1
1
1
b
=>
a
20V
1Ω
2/3
1Ω
10V
b
=>
a
2/3
b
10V
2Ω
=>
a
2/3 Ω
b
2Ω
5A
a
1/2
b
a
1/2 Ω
b
2.5

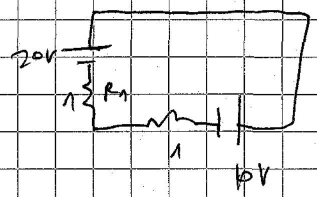
בדק
-1

②

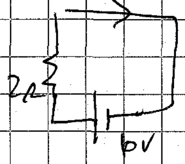




נניח שד1 וד2 במגמה קדמית


20V
1
R1
1
10V

→



הזרם בכיוון הנכון
כלומר שהנחה
של ה' טובה.

כיון צריך 10
הולך המעגל.
ולא זה שקול
תמיד.

-2

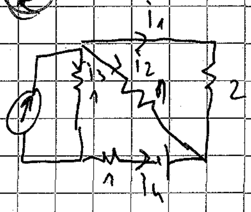
1~~00~~

~~[illegible]~~

ולכן עובר ב 1 זרם של 5A

$$P = I^2R = 25 \cdot 1 = \underline{25\,W}.$$

③


i_1
i_2
i_3
1
1
2
1
i_4

$i_1 + i_2 + i_3 = 20_A$

$2i_1 = i_2$

$3i_1 + i_3 = 20$

$i_3 = 20 - 3i_1$

$i_2 + 10 + i_1 + i_2 - i_3 = 0$

$2i_1 + 10 + i_1 + 2i_1 - 20 + 3i_1 = 0$

$10 = -10 + 5 + -15$

$2i_1 = 10$

$8i_1 = 10$

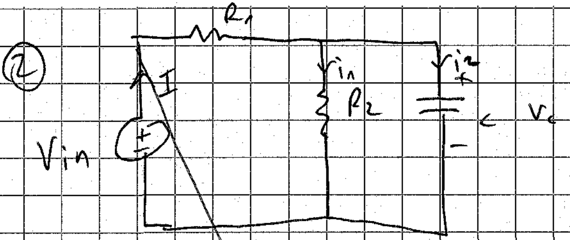
$$\begin{cases} i_1 = \frac{10}{8} \\ i_2 = \frac{20}{8} \\ i_3 = 20 - \frac{30}{8} \\ \quad i_3 = 16.25 \\ i_4 = -i_1 - i_2 = -\frac{30}{8} \end{cases}$$

$$\begin{cases} i_1 = \cancel{10}\ 5_A \\ i_2 = 10_A \\ i_3 = 5_A \\ i_4 = -i_1 - i_2 = -15_A \end{cases}$$

✓

ב׳

②



ⓐ

$$i_1 \cdot R_2 = \underset{\overset{\|}{0}}{V_{c(0)}} + \frac{1}{C}\int i_2\, d\tau = V_C$$

$$i_1 = \frac{\dot V_C}{R_2} \qquad i_2 = C\dot V_c$$

$$i_1 = \frac{V_c}{R_2}$$

$$V_{in} = R_1\cdot(i_1+i_2) + V_c = R_1\left(\frac{V_c}{R_2} + C\dot V_c\right) + V_c$$

$$\dot V_c + \frac{R_1+R_2}{CR_2R_1}V_C = \frac{V_{in}}{R_1C}$$

$$R_1C\,\dot V_c + \left(\frac{R_1}{R_2}+1\right)V_c = V_{in} \Rightarrow$$

ZIR = 0 , תנאי התחלה 0

עירור $V_{in} = U(t)$

$$\cancel{\left(\frac{R_1}{R_2}+1\right)V_c = 1}$$

$$\frac{R_1+R_2}{\cancel{C}R_2\cancel{R_1}}V_{c(p)} = \frac{1}{\cancel{R_1C}}$$

$$V_{cp} = \frac{R_2}{R_1+R_2}$$

$$V_c = A\cdot e^{-\frac{R_1+R_2}{CR_1R_2}t} + \frac{R_2}{R_1+R_2}$$

$$V_c(0) = 0 \qquad A = -\frac{R_2}{R_1+R_2}$$

ZSR: $V_{c}(t) = \frac{R_2}{R_1+R_2}\left(1 - e^{-\frac{R_1+R_2}{CR_1R_2}t}\right)\cdot U(t)$

ⓑ

$t\to\infty$

$V_{in} - V_c$ על R_1 (מתח)

$$V_c(t\to\infty) = \frac{R_2}{R_1+R_2} \qquad V_{R_1} = \frac{R_1+R_2-R_2}{R_1+R_2} = \frac{R_1}{R_1+R_2}$$

הערות
הבודק

ג-7

$R_1 = 1\Omega \quad ; \quad R_2 = k\Omega$

$0-20\ kHz$

$$H(\omega) = \left|\frac{V_{out}}{V_{in}}\right| \geq \frac{1}{\sqrt{2}}\left|H(\omega=0)\right|$$

אני מניח שמשוואת התדרים ת-ל רצפה לכן מדרגה..

ולאחר מכן אין צורך בנגד עזרי

ולעזרה התנגדות הכל 0

0 אין נתונה - הגבלה מחשב, אבל לגבי סוגים שונים:

$$\hat{V}_{in} = (\hat{I}_1 + \hat{I}_2) R_1 + \hat{I}_1 R_2$$

$$\hat{I}_1 R_2 = \frac{\hat{I}_2}{j\omega C}$$

$$\hat{I}_1 = \frac{\hat{I}_2}{jR_2\omega C}$$

$$\hat{V} = \hat{I}_2\left(\frac{jR_1R_2\omega C + R_1}{jR_2\omega C}\right) + \hat{I}_2\frac{R_2}{jR_2\omega C}$$

$$\hat{V}_{in} = \hat{I}_2\left(\frac{jR_1R_2\omega C + R_1 + R_2}{jR_2\omega C}\right)$$

$$V_c = \frac{V_{in}\cdot jR_2\cancel{\omega C}}{(jR_1R_2\omega C + R_1 + R_2)\, j\cancel{\omega C}} =$$

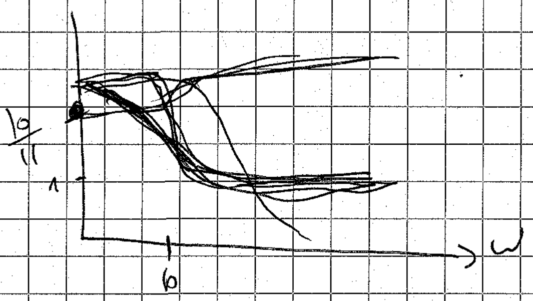
$$H(\omega) = \frac{V_{out}}{V_{in}} = \frac{R_2}{jR_1R_2\omega C + R_1 + R_2} \geq \frac{1}{\sqrt{2}}\ \frac{R_2}{R_1 + R_2}$$

$$\frac{j\cancel{R_1R_2}\omega C + R_1 + R_2}{R_1 + R_2} \leq 1\ \cancel{R_2}$$

$$jR_1R_2\omega C + R_1 + R_2 \leq \sqrt{2}R_1 + \sqrt{2}R_2$$

$$jR_1R_2\omega C \leq (R_1 + R_2)(\sqrt{2} - 1) \qquad \cancel{\omega} \leq \frac{R_1 + R_2(\sqrt{2}-1)}{jR_1R_2\omega}$$

טיוטה למבחן



$$\left|\frac{10\omega jc+110}{11+10\omega jc}\right| \geq \frac{10}{11\sqrt{2}}$$

$$\left|\frac{(10\omega jc+110)(11-10\omega jc)}{11^2+100\omega^2c^2}\right| \geq \frac{10}{11\sqrt{2}}$$

$$\left|\frac{110\omega jc-100\omega jc+110+100\omega^2c^2}{11^2+100\omega^2c^2}\right| \overset{?}{\geq} \frac{10}{11\sqrt{2}}$$

$$\frac{100\omega^2c^2+(110+100\omega^2c^2)^2}{(11^2+100\omega^2c^2)^2} \geq \frac{100}{2\cdot 11^2}$$

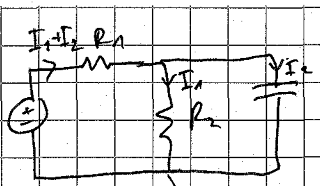
$$100\omega^2c^2+12100+22000\omega^2c^2+10000\omega^4c^4 \geq \frac{100}{242}($$

$$14641+24200\omega^2c^2+10000\omega^4c^2$$

$$\frac{99}{121}\omega^4c^4 \; 2100\,\omega^2c^4-2541 \geq 0$$

הערות
הבודק

ד'



$$\hat{V} = (\hat{I}_1 + \hat{I}_2) R_1 + \hat{I}_1 R_2$$

$$\hat{I}_1 R_2 = \frac{\hat{I}_2}{j\omega c} \qquad \hat{I}_1 = \frac{\hat{I}_2}{R_2 \omega c j}$$

$$\hat{V} = \hat{I}_2 \left(R_1 + \frac{R_1}{R_2 \omega c j} + \frac{1}{\omega c j} \right) = I_2 \left(\frac{R_1 R_2 \omega c j + R_1 + R_2}{R_2 \omega c j} \right)$$

$$I_2 = \frac{\hat{V}_{in} R_2 \omega j c}{R_1 + R_2 + R_1 R_2 \omega c j}$$

②

$$H(\omega) \left| \frac{V_c}{V_{in}} \right| = \frac{R_2 \omega j c}{R_1 + R_2 + R_1 R_2 \omega c j} \overset{?}{\geq} \frac{1}{\sqrt{2}} = \frac{R_2 \omega j c (R_1 + R_2 - R_1 R_2 \omega c j)}{(R_1 + R_2)^2 + (R_1 R_2 \omega c)^2}$$

$$\geq \frac{1}{\sqrt{2}}$$

$$\sim\!\!\sim (R_2 + R_1) R_2 \omega j c + R_2^2 R_1 c^2 \omega^2$$

$$H(\omega) \frac{V_{R_1}}{V_{in}} = \frac{(I_1 + I_2) R_1}{V_{in}} = R_1 \left(\frac{R_2 \omega j c + R_2}{R_1 + R_2 + R_1 R_2 \omega j c} \right) \frac{V_{in}}{V_{in}}$$

ד. ועכשיו ל- V_{R1} ו- V_c אותה [illegible] G ה[illegible]

לא יוצא...

$$H(\omega) = \frac{R_2 \omega j c + R_2}{R_1 + R_2 + R_1 R_2 \omega j c} \geq \frac{1}{\sqrt{2}} \frac{R_2}{R_1 + R_2}$$

$$\left| \frac{10 \omega j c + 10}{11 + 10 \omega j c} \right| \geq \frac{1}{\sqrt{2}} \frac{10}{11}$$

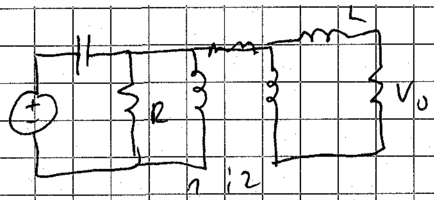
$$10 \omega j c + 10 \geq \frac{10}{11\sqrt{2}} (11 + 10 \omega j c)$$

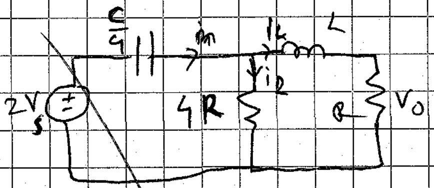
$$10 \omega j c - \frac{100 \omega j c}{11\sqrt{2}} \geq \frac{10}{\sqrt{2}} - 10$$

$$\sqrt{\frac{(10 \omega j c + 10)(11 - 10 \omega j c)}{11^2 + 100 \omega^2 c^2}} \geq \frac{10}{11\sqrt{2}} \Rightarrow 110 \omega j c$$

הערות
הבודק

②





$$\frac{4}{C}\int I_L + i_2 + 4R i_2 = 2V_s$$

$$\frac{4}{C}\left(I_L + i_2\right) + 4R i_2' = 2\dot{V}_s$$

$$L i_L' + i_L R = i_2 R \cdot 4$$

$$\frac{4}{C}\left(I_L + \frac{I_L}{4} + \frac{i_L'}{4R}\right) + 4R I_L' = 2\dot{V}_s$$

$$i_2 = \frac{L i_L'}{4R} + \frac{i_L}{4}$$

$$\frac{4V_R}{RC} + \frac{V_R}{RC} + \left(\frac{L}{CR^2} + 4\right)V_R' = 2RV_s'$$

$$\frac{4}{C}\left(\frac{5}{4}I_L + \frac{L I_L'}{4R}\right) + 4R\left(\frac{L I_L''}{4R} + \frac{I_L'}{4}\right) = 2V_s'$$

$$\frac{5}{RC}(RI_L) + \frac{L}{C4R^2}(RI_L') + \frac{L}{R}(RI_L'') + RI_L' = 2V_s'$$

$$\frac{5}{RC}V_R + \left(1 + \frac{L}{4R^2C}\right)V_R' + \frac{L}{R}V_R'' = 2V_s' \quad / \cdot \frac{R}{L}$$

$$V_R'' + \left(\frac{R}{L} + \frac{1}{4RC}\right)V_R' + \frac{5}{LC}V_R = \frac{2R}{L}V_s'$$

$$2V_s = \frac{4}{C}\int i_L + i_R + 4R i_R$$

$$2\dot{V}_s = \frac{4}{C}i_L + \frac{4L}{C}\frac{I_L'}{4R} + \frac{4}{C}\frac{I_L}{4} + 4RL\frac{I_L''}{4R} + \frac{4R\,i_L'}{4}$$

$$2\dot{V}_s = \frac{5}{C}i_L + i_L'\left(\frac{L}{RC} + R\right) + L I_L'' \quad / \cdot \frac{R}{L}$$

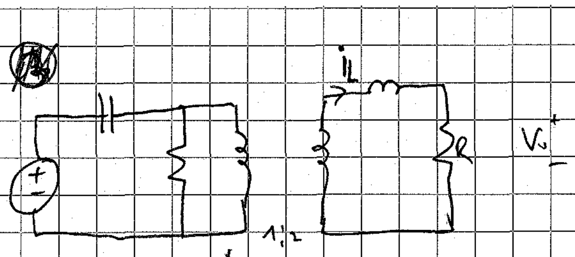
$$\frac{2R}{L}\dot{V}_s = \frac{5}{CL}V_c + V_c'\left(\frac{1}{RC} + \frac{R}{L}\right) + V_R''$$

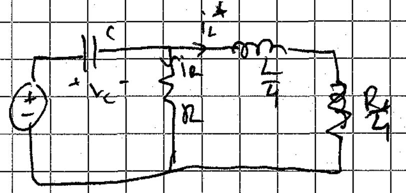
[illegible]

הערות
הבודק

~~[illegible]~~

③





$$V_S = \int (i_L^* + i_R)\,dt \cdot \frac{1}{C} + i_R \cdot R$$

$$i_R \cdot R = i_L^{*\prime} \frac{L}{4} + i_L^* \cdot \frac{R}{4}$$

$$i_R = i_L^{*\prime} \cdot \frac{L}{4R} + \frac{i_L^*}{4}$$

$$\dot{V}_S = \frac{1}{C}(i_L^* + i_R) + i_R' \cdot R$$

$$\dot{V}_S = \frac{1}{C}\left(i_L^* + \dot{i}_L^* \cdot \frac{L}{4R} + i_L^* \frac{\cancel{R}\,1}{4}\right) + \cancel{R}\frac{i_L^{\prime\prime *}}{4\cancel{R}} + \frac{i_L^{\prime *}}{4} \cdot R$$

$$\dot{V}_L^* \cdot R = V_R \qquad i_L^* = \frac{V_R}{R}$$

~~[illegible]~~

$$\dot{V}_S = \frac{1}{C}\left(\frac{V_R}{R} + \frac{\dot{V}_R L}{4R^2} + \frac{V_R}{4R}\right) + \frac{\ddot{V}_R}{4R} + V_R \;/\cdot 4R.$$

$$\ddot{V}_R + \frac{L}{R}\dot{V}_R + \left(\frac{5}{C} + 4R\right)V_R = 4R\dot{V}_S$$

טיוטה למבחן

$$I_2 = I_1' \frac{L}{4R} + \frac{I_1}{4}$$

$$I_2' = I_1'' \frac{L}{4R} + \frac{I_1'}{4}$$

$$\dot{V}_s = \frac{1}{C}\left(\frac{5}{4}I_1 + \frac{L}{4R}I_1'\right) + I_1''\frac{L\cdot\cancel{R}}{4\cancel{R}} + \frac{I_1'}{4}\cdot R$$

$$\dot{V}_s = \frac{5}{4C}I_1 + \left(\frac{L}{4RC} + \frac{R}{4}\right)I_1' + \frac{L}{4}I_1'' \quad \Big/\cdot\frac{4R}{L}$$

$$\frac{4R}{L}\dot{V}_s = \frac{5}{LC}V_R + \left(\frac{1}{RC} + \frac{R}{L}\right)V_R' + V_R''$$

V_o

$I_L = A$

$$V_o = I_L \cdot R = RA$$

I_L

$V_C = B$

$$\dot{i}_L' \cdot L + V_o = 2V_s - 2V_c$$

$$i_L' \cdot L = 2V_s - 2B - RA$$

$$i_L' = \frac{2V_s - 2B - RA}{L}$$

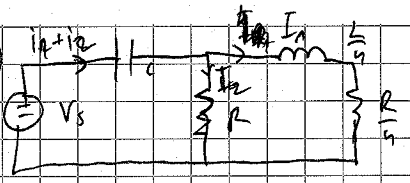
$$R i_L' = V_o' = \frac{R}{L}(2V_s - 2B - RA)$$

$$V_R'' + \left(\frac{1}{RC} + \frac{R}{L}\right)V_R' + \frac{5}{LC} = \frac{2R}{L}\dot{V}_s$$

(ב) יש שקול, כי שינוי רק [illegible] הגורמים [illegible]

המדר [illegible]) V_o הזה

$$\boxed{V_R'' + \left(\frac{1}{RC} + \frac{R}{L}\right)V_R' + \frac{5}{CL}V_R = 2\frac{R}{L}V_s'}$$

③ 

$1:2$

כ"פ

$V_0 = 2V_{0\,(real)}$

~~$\dot{i}_R\frac{L}{4} + \dot{i}_R\cdot\frac{R}{4} = \frac{1}{C}$~~

~~$\frac{1}{C}(i_L + i_R) + \frac{1}{C}$~~

$V_0 + V_C = V_s$

$V_0 = V_s - V_c$

$\dot{V}_s = \frac{1}{C}(I_1 + I_2) + \dot{I}_2 R$

$I_2 \cdot R = I_1'\frac{L}{4} + I_1\frac{R}{4}$

$I_2 = I_1'\frac{L}{4R} + \frac{I_1}{4}$ $\qquad$ $\dot{I}_2 = I_1''\frac{L}{4R} + \frac{I_1'}{4}$

$\dot{V}_S = \frac{1}{C}\left(I_1 + I_1'\frac{L}{4R} + \frac{I_1}{4}\right) + I_1''\frac{L}{4} + I_1'\frac{R}{4} \quad /\cdot\frac{R}{4}$

$I_1\cdot\frac{R}{4} = V_0$

$\frac{R}{4}\dot{V}_s = \frac{R}{4C}I_1 + \frac{R}{4}I_1\cdot\frac{1}{4C} + \frac{R}{16}I_1'\cdot\frac{L}{4R} + I_1''\frac{R}{4}\cdot\frac{L}{4} + I_1'\frac{R}{4}$

$\frac{R}{4}\dot{V}_s = V_0\left(\frac{1}{C} + \frac{1}{4C}\right) + \boxed{V_0'\frac{L}{4R}} + V_0''\frac{L}{4} + V_0\frac{R}{4} \quad /\cdot\frac{4}{L}$

$$\boxed{V_0'' + \left(\frac{1}{RC} + \frac{R}{L}\right)V_0' + \left(\frac{4}{LC} + \frac{1}{LC}\right)V_0 = \left(\frac{R}{L}V_s'\right) \Rightarrow \frac{2R}{L}V_s'^{*}}$$

[illegible] $\leftarrow V_0^* = 2V_0 \rightarrow$ ([illegible])

ב $V_C(0^-) = B$ $\qquad$ $V_0(0^-) = I_L(0^-)\cdot R$ | V_0^* [illegible] V_0 [illegible]

$I_L(0^-) = A$

$$\boxed{\begin{cases} V_0(0^-) = AR \\ V_0'(0^-) = \frac{1}{L}(2RV_s - 2RB - AR^2) \end{cases}}$$

$V_0'(0) = i_L'\cdot R$

$i_L'\cdot L + i_L R =$ ([illegible])

$V_0 = 2V_s - 2V_c$ $\qquad$ $\overset{V_0(0^-)}{i_L'\cdot L + i_L R} = 2V_s - 2V_c$

~~$i_L = \frac{2V_s - 2V_c}{R}$~~ $\qquad$ $i_L' = \frac{2V_s - 2V_c - AR}{L}$

$V_c = \int i_L + i_2$ $\qquad$ $V_0' = i_L'\cdot R$

הערות
הבודק

3.0ד

$V_0'' + 2V_0' + 5V_0 = 2V_s$

$V_0(0^-) = 1v$

$V_0'(0^-) = 3\ \frac{volt}{sec}$

② ZIR :

$\lambda^2 + 2\lambda + 5 = 0 \qquad \frac{-2 \pm \sqrt{4 - 4\cdot1\cdot5}}{2} \qquad \frac{-2 \pm 4i}{2}$

$S_1 = -1 + 2i \qquad S_2 = -1 - 2i$

$V_0 = e^{-t}(A\sin(2t) + B\cos(2t))$

$V(t=0) = B = 1$

$V_0'(t=0) = -e^{-t}(\cos(2t)) + e^{-t}(2A) = 3$

$-1 + 2A = 3 \qquad 2A = 4 \qquad A = 2$ ✓

$V_{0\,ZIR} = e^{-t}(2\sin 2t + \cos 2t)$

③ $V(0^-) = 0$ אבל ברגע, [illegible]

$V'(0^-) = 2$

$B = 0$

$V_0' = e^{-t}\, 2A\cos(2t) = 2$ ✓

$A = 1.$

$V_{0\,ZSR} = (e^{-t}\sin 2t)\cdot u(t)$

סה"כ:

$V_0(t) = e^{-t}(2\sin 2t + \cos 2t) + (e^{-t}\sin 2t)\cdot u(t)$

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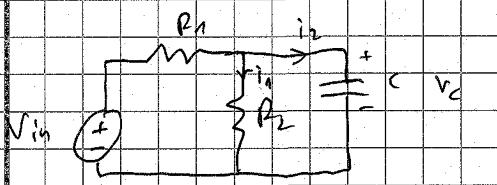
הערות
הבודק

②

מספר שאלה ← שאלה 2:

31|33



א) $C\dot{V}_c = i_2$

$$i_1 \cdot R_2 = V_c$$

$$i_1 = \frac{V_c}{R_2}$$

$$V_s = \left(\frac{V_c}{R_2} + C\dot{V}_c\right)\cdot R_1 + V_c$$

$$V_s = R_1 C\,\dot{V}_c + \left(1+\frac{R_1}{R_2}\right)V_c$$

$$\dot{V}_c + \frac{R_2+R_1}{R_1R_2C}V_c = \frac{V_{in}}{R_1C}$$ ✓ גמ"ר!

$$V_c(t) = \underbrace{A\cdot e^{-\frac{R_1+R_2}{CR_1R_2}t}}_{V_H} + V_{cp}$$

$$\frac{R_2+R_1}{R_1R_2C}V_{cp} = \frac{1}{R_1C} \qquad V_{cp} = \frac{\cancel{R_1}R_2\cancel{C}}{R_1+R_2} = \frac{R_2}{R_1+R_2}$$

$$A\cdot e^{\frac{-R_1+R_2}{CR_1R_2}t} + \frac{R_2}{R_1+R_2} = 0$$

$$A = -\frac{R_2}{R_1+R_2}$$

[illegible] $V_c = \frac{R_2}{R_1+R_2}\left(1-e^{-\frac{R_1+R_2}{CR_1R_2}}\right)\cdot U(t)$ ✓ תגובה למדרגה

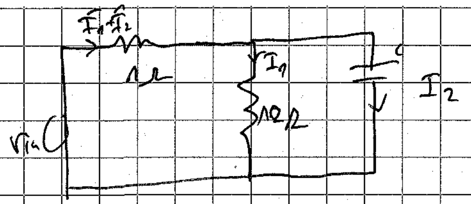
[illegible]

ב) $V_R(t\to\infty) = V_{in} - V_c(t\to\infty) = 1 - \frac{R_2}{R_1+R_2} = \frac{R_1}{R_1+R_2}$

✓

הערות הבודק

②



$$10 I_1 = \frac{I_2}{j\omega c} \qquad I_1 = \frac{I_2}{10 j\omega c}$$

$$\hat V = \hat I_1 + \hat I_2 + 10 I_1 = 11 I_1 + I_2 = I_2\left(\frac{11}{10 j\omega c}+1\right)$$

$$\hat V = \hat I_2\left(\frac{11+10j\omega c}{10 j\omega c}\right) = I_2\left(\frac{-110 j\omega c + 100\omega^2 c^2}{100\omega^2 c^2}\right)$$

$$\hat I_2 = \frac{\hat V\, 100\,\omega^2 c^2}{100\omega^2 c^2 - 110\omega j c} \qquad \hat I_2 = \frac{\hat V\, 10 j\omega c}{11+10j\omega c}$$

$$\hat I_1 = \frac{\hat V}{11+10j\omega c} \quad ✓$$

$$\hat V_C = \hat I_1 \cdot R_2 = \frac{10\,\hat V}{11+10j\omega c} \qquad (V_c \parallel V_{R_2})$$

$$H(\omega) = \left|\frac{V_{out}}{V_{in}}\right| = \left|\frac{10}{11+10j\omega c}\right| = \frac{10}{\sqrt{121+100\omega^2 c^2}}$$

$$H(\omega) \geq \frac{1}{\sqrt2}\left|H(\omega=0)\right|$$

$$H(\omega) \geq \frac{1}{\sqrt2}\cdot\frac{10}{11}$$

$$\frac{\cancel{10}}{\sqrt{121+100\omega^2c^2}} \geq \frac{\cancel{10}}{11\sqrt2} \qquad (11\sqrt2)^2 \geq 121+100\omega^2c^2$$

$$242 \geq 121+100\omega^2c^2$$

בתדר השינוי ↓

$$121 \geq 100\,\omega^2 c^2$$

$$0<\omega^2<(20\cdot10^3)^2(2\pi)^2 = 1.579\cdot10^{10} \qquad \omega^2c^2 \leq 1.21$$

$\omega = 2\pi f$ $\qquad 0<c^2<$ ~~8.753 μF~~

$f = 20\cdot10^3$ $\qquad 0<c^2<7.66\cdot10^{-11}$

~~כדי ש [illegible] התדרים [illegible] ω~~ ← $\boxed{0<c<8.753\,\mu F}$ ⇐ ✓

⇐ נשים ב V_{out} ~~[illegible]~~ מתח על הקבל (או על הנגד R_2)

(כדי ש[illegible] עובר רק התדרים הנמוכים [illegible] שישמשו [illegible]

ונשים לב שבגלל ש הזרם [illegible]) מתח.

נראה מהגרף (כנראה) [illegible] ש [illegible] V_{out} מ R_1 - לא נעדיף דווקא ערכים [illegible] לכן זה טובה.

הערות הבודק

ב'

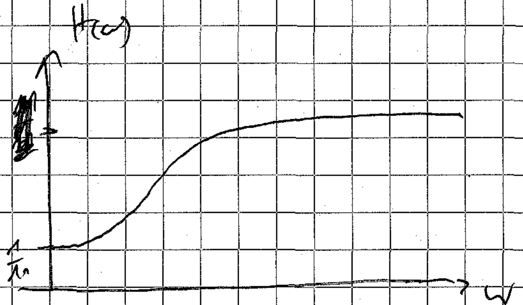
③

$$H(\omega) = \frac{V_{R_1}}{V_{in}} = \frac{(I_2 + I_1)\cdot R_1}{V_{in}}$$

$$H(\omega) = \frac{\hat{V}}{\hat{V}}\left(\frac{10j\omega C + 1}{11 + 10j\omega C}\right)\cdot \underset{''}{R_1}$$

$$\omega \to 0 \quad H(\omega) = \frac{1}{11}$$

$$\omega \to \infty \quad H(\omega) \to 1$$



✓

אני לא כ"כ יודע איך לסכם בשלב זה.

~~[illegible]~~ ו'.

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